

AIRBNB INSIGHTS

Customer Segmentation,
Product Offerings, and
Pricing Strategies

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THE CHALLENGE

Many Airbnb hosts and investors struggle with:

- Identifying the right target customers
- Deciding on the right property and amenities
- Setting competitive prices

Result: Missed revenue opportunities and low occupancy

To address the challenge, this project aims to:

- Segment guest behaviors and preferences from Airbnb reviews using Natural Language Processing algorithms to identify customer personas
- Conduct a comprehensive data analysis to recommend product offerings and pricing strategies for each customer personas

THE DATASET

- Source from “Inside Airbnb” a public data project that provides detailed and downloadable datasets about Airbnb listings in major cities worldwide.
- Bangkok, Central Thailand, Thailand dataset scraped on September 26, 2025
- Listings
 - 28,806 Airbnb listings with 79 columns
- Reviews
 - 583, 333 guest reviews from 18,719 listings

TARGET CUSTOMERS



CUSTOMER SEGMENTATION

Topic modeling with **Latent Dirichlet Allocation** is used to analyze guest reviews and identify common topics behind guests' motivation for booking an Airbnb. This algorithm groups similar words to reveal what guests consistently value.

Approach:

- Identified key topics from text reviews
- Focused on guest behaviors and preferences
- Refined results into clear customer personas

Key Customer Personas:

- Home Comfort Seekers
- Local Transit Explorers

TARGET CUSTOMERS

Customer Persona 1: Home Comfort Seekers

['clean', 'room', 'comfortable', 'pool', 'home', 'bed', 'view', 'gym', 'spacious', 'big']

- Travelers focused on relaxation and indoor comfort. They prefer clean, spacious, and well-maintained stays with a calm and private atmosphere. Their decisions are driven by comfort quality, aesthetic appeal, and overall stay experience, with strong attention to visual presentation and guest reviews.

Sample review representing this persona

“We highly recommend this condo room for its cleanliness, set up, and hospitality. We enjoyed our experience from the very start. The host was very accommodating to our needs, and a good communicator who responded quickly with perfectly spoken English.

The condo room was very clean, and very well equipped (Plates, cutlery etc) for 2 adults, which included a small (but very suitable) dining table with 2 chairs. Located on the 16th floor, the room offers a good night time view of the city, and has very quiet neighbours. The T.V. is bigger than most airbnb rooms, had English speaking cable movies, and I was able to hook my laptop up and view my Netflix from the very comfortable leather lounge.

We'll definitely check the availability of this condo room when we return. Thank you.”

TARGET CUSTOMERS

Customer Persona 2: Local Transit Explorers

['walk', 'bts', 'station', 'restaurant', 'food', 'close', 'minute', 'area', 'local', 'distance']

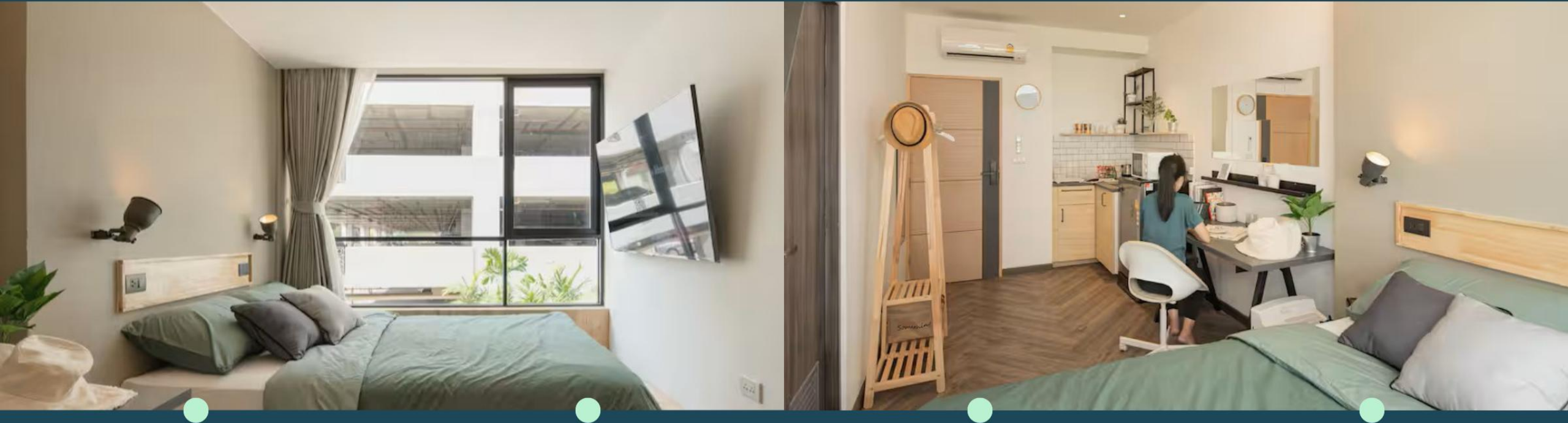
- Travelers who prioritize convenience and accessibility. They prefer well-located stays that make it easy to move around, explore nearby areas, and access local food and attractions. Their choices are driven by practicality, location efficiency, and ease of travel.

Sample review representing this persona

“EXCELLENT LOCATION. Close to the BTS train, which you can take to reach the Chao Praya River in just 5 minutes. From Sathor Pier, you can take a very affordable boat to Bangkok's major tourist attractions, such as the Grand Palace, Wat Aron, and others.

LOCAL RESTAURANTS (local market): 300 meters from the apartment. In this local market you can find more than 30 restaurants with local food, and also fruits and other goods, half of the price that you usually pay in food courts at Malls. If you are at the apartment search for Madame Beef and Google Maps will take you to a clean local market.”

PRODUCT OFFERINGS

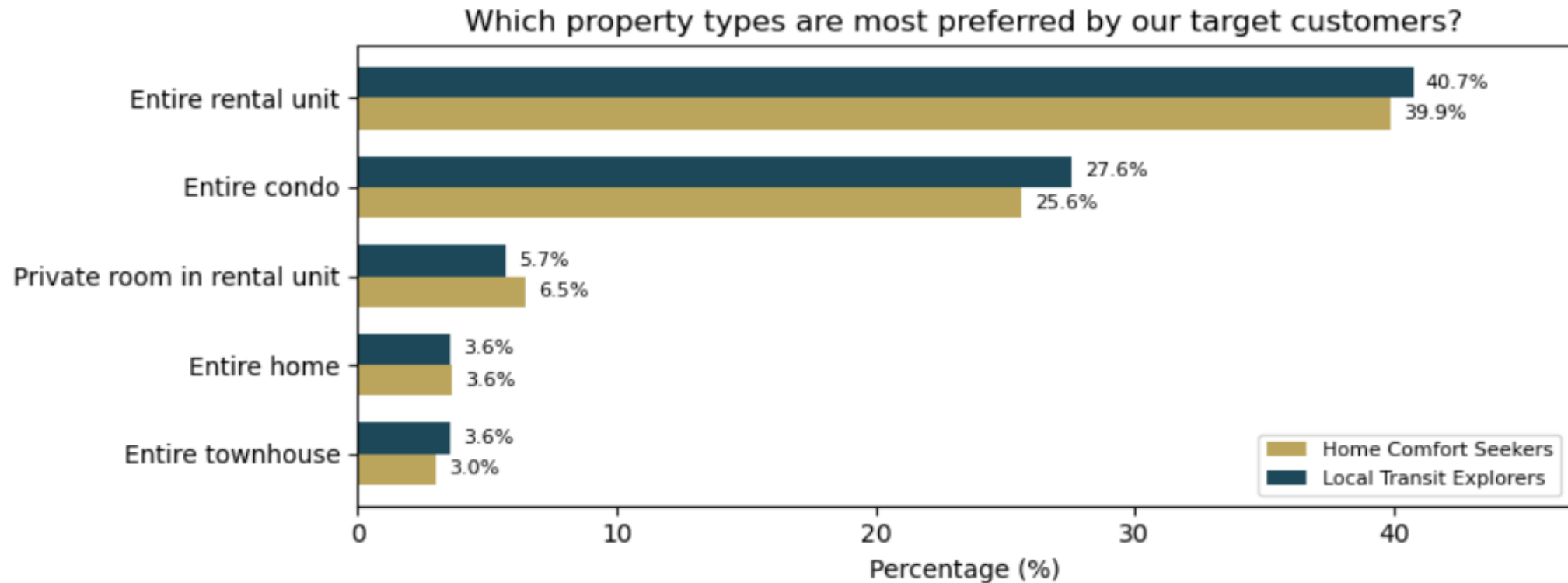


PRODUCT OFFERINGS

Exploratory data analysis is conducted to understand the expressed preferences of each target customer segment. This section identifies the most common features guests look for when booking an Airbnb, specifically addressing the following key questions:

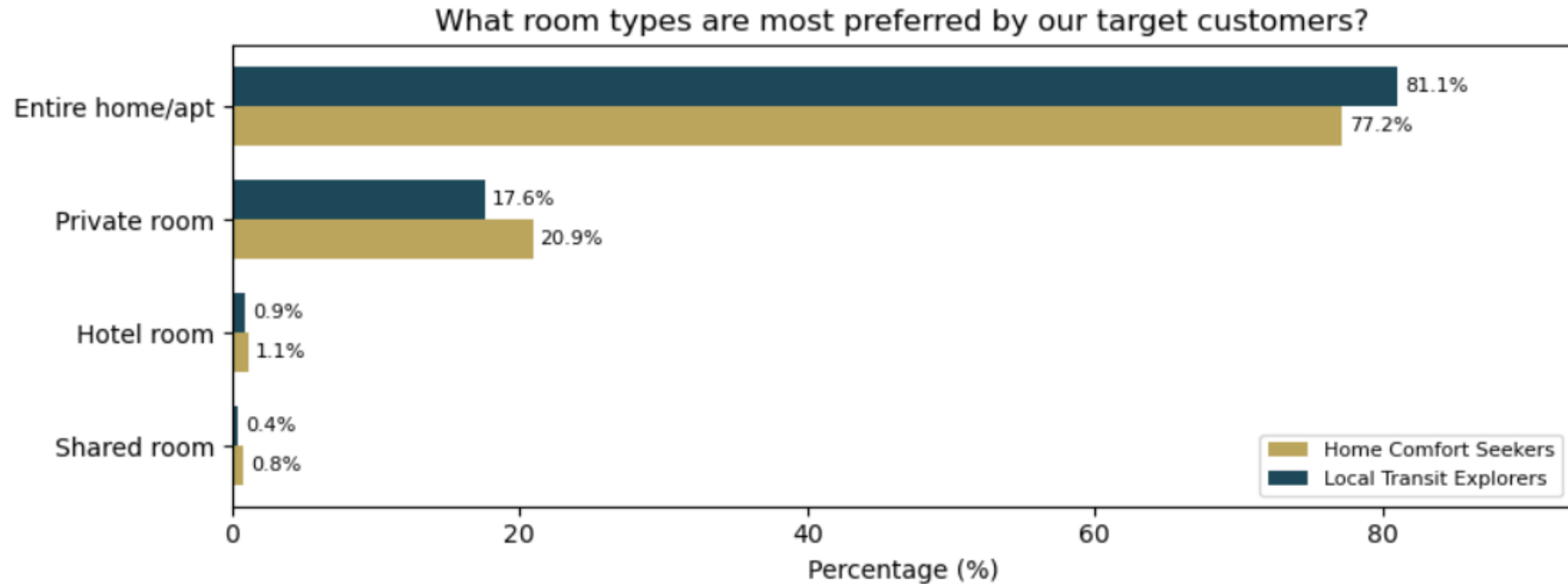
- *Which property types are most preferred by our target customers?*
- *What room types are most preferred by our target customers?*
- *What group sizes do target customers usually book for?*
- *How many bedrooms are preferred by our target customers?*
- *How many beds do target customers actually need per stay?*
- *How many bathrooms do target customers prefer?*
- *Are preferred bathrooms shared or private?*
- *What is the most commonly expected amenity among Home Comfort Seekers?*
- *What is the most commonly expected amenity among Local Transit Explorers?*

PROPERTY TYPE



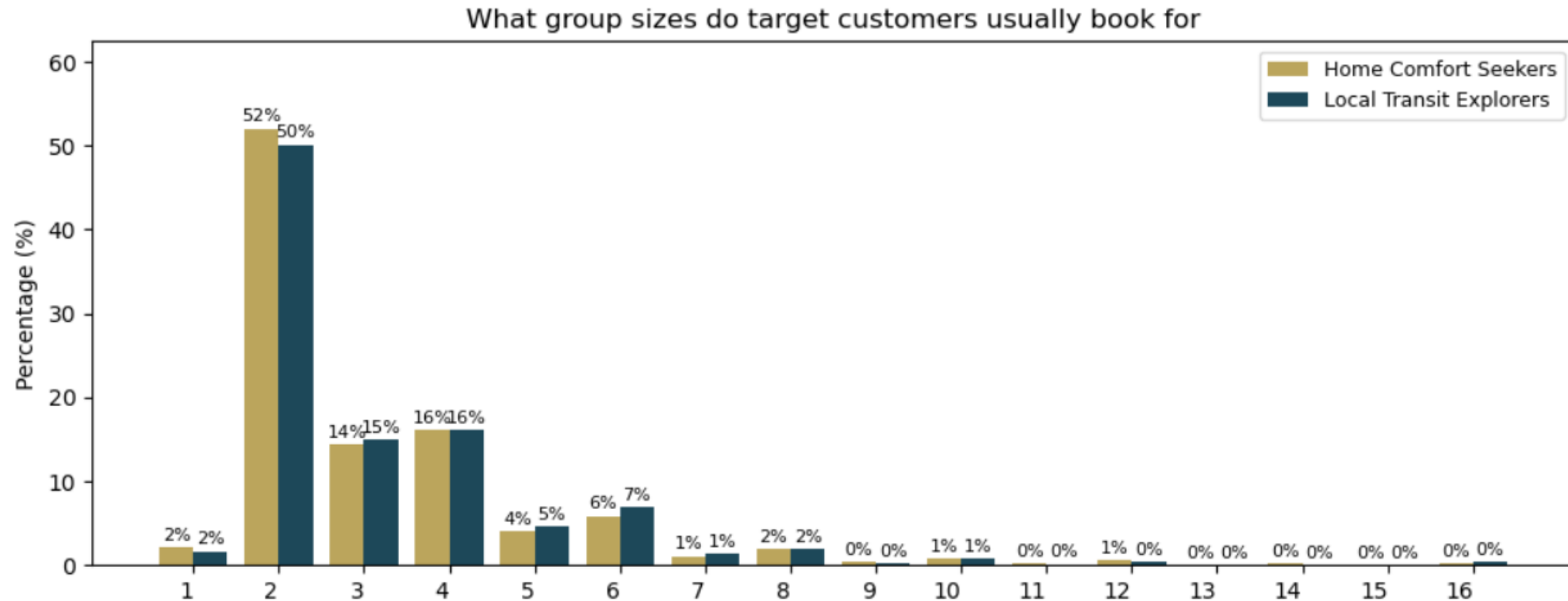
Demand is heavily concentrated in fully private accommodations, led by entire rental units and entire condos across both segments. Shared accommodations (private rooms) show consistently low preference, reinforcing a strong bias toward privacy.

ROOM TYPE



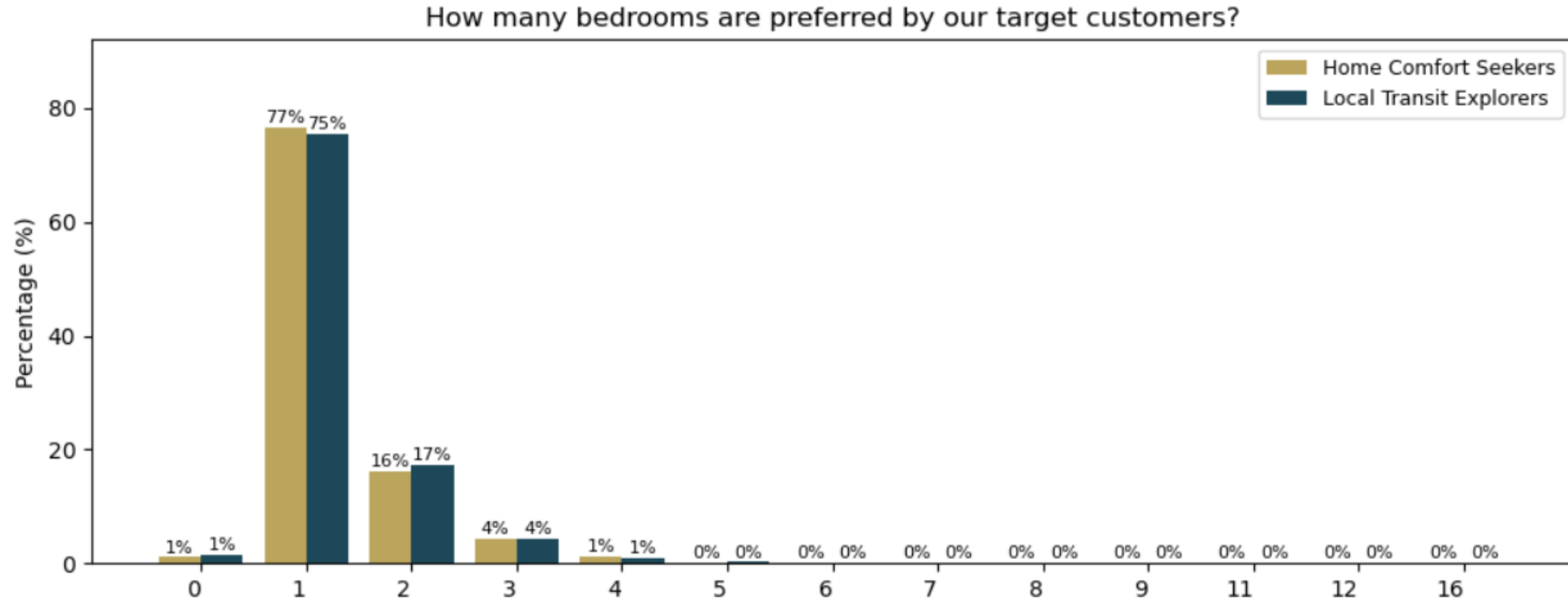
Most customers prefer entire homes or apartments, with demand far higher than private rooms. This shows a clear preference for privacy, so listings should focus on fully private stays to match customer demand.

GROUP SIZE



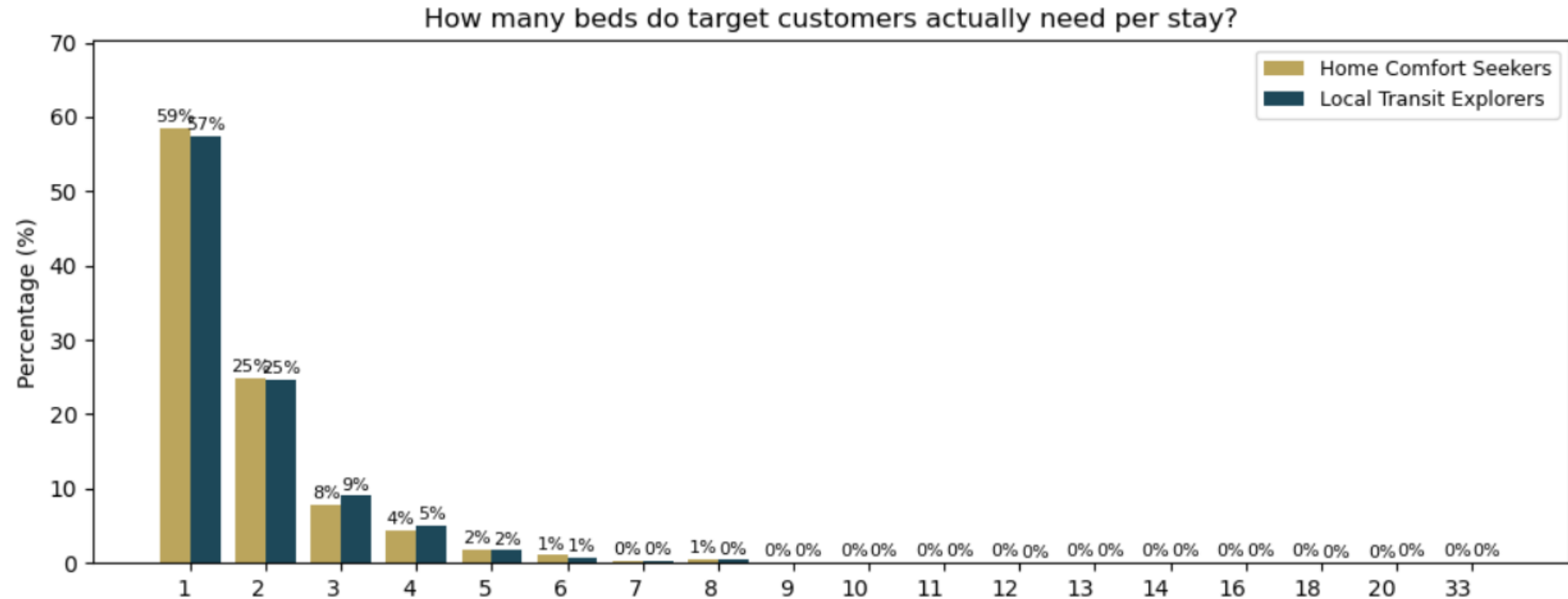
Most bookings are for 2 people, with consistent demand for groups of up to 4. This suggests prioritizing units for 2 guests while maintaining flexibility to accommodate up to 4 when needed.

BEDROOMS



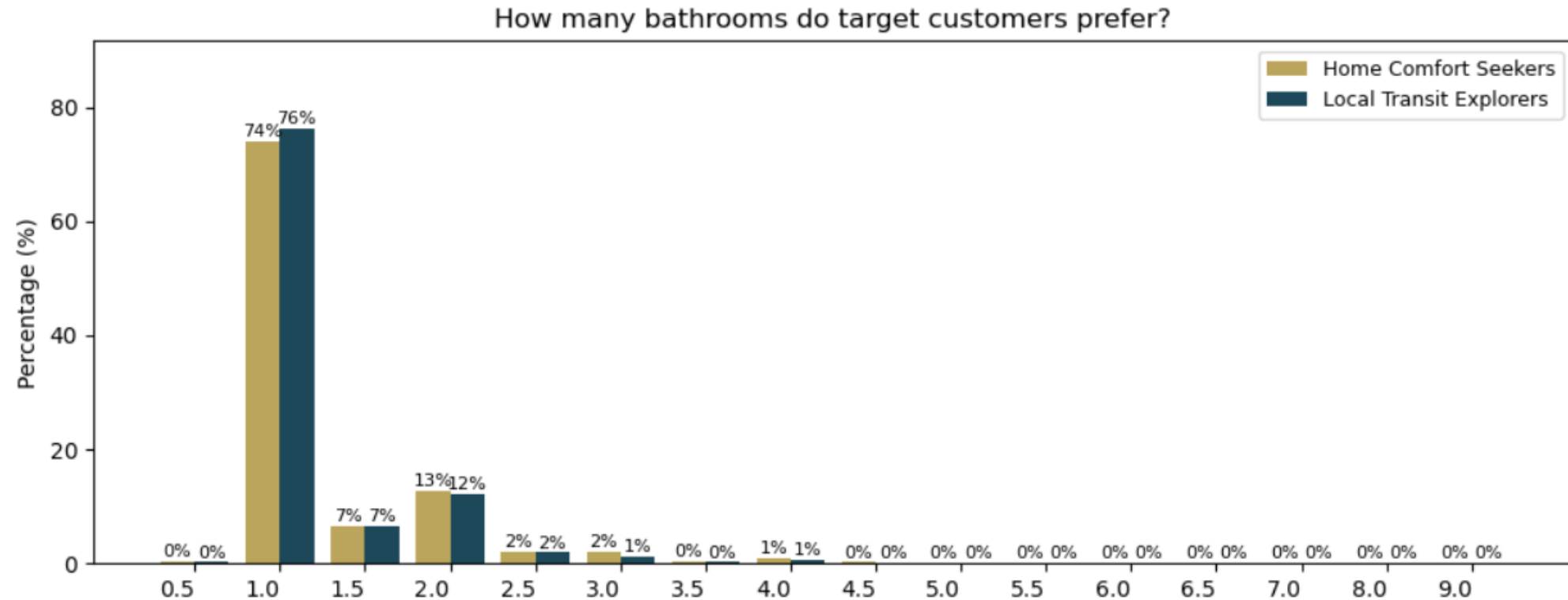
Most customers strongly prefer 1-bedroom units, far above 2-bedroom units, which are a secondary choice. This shows demand is concentrated in smaller units, so listings should prioritize 1-bedroom properties.

BEDS



Most customers prefer listings with 1 bed, followed by 2 beds as a secondary option. This shows strong demand for simple bed setups, so listings should mainly offer single-bed configurations with some flexibility for two beds.

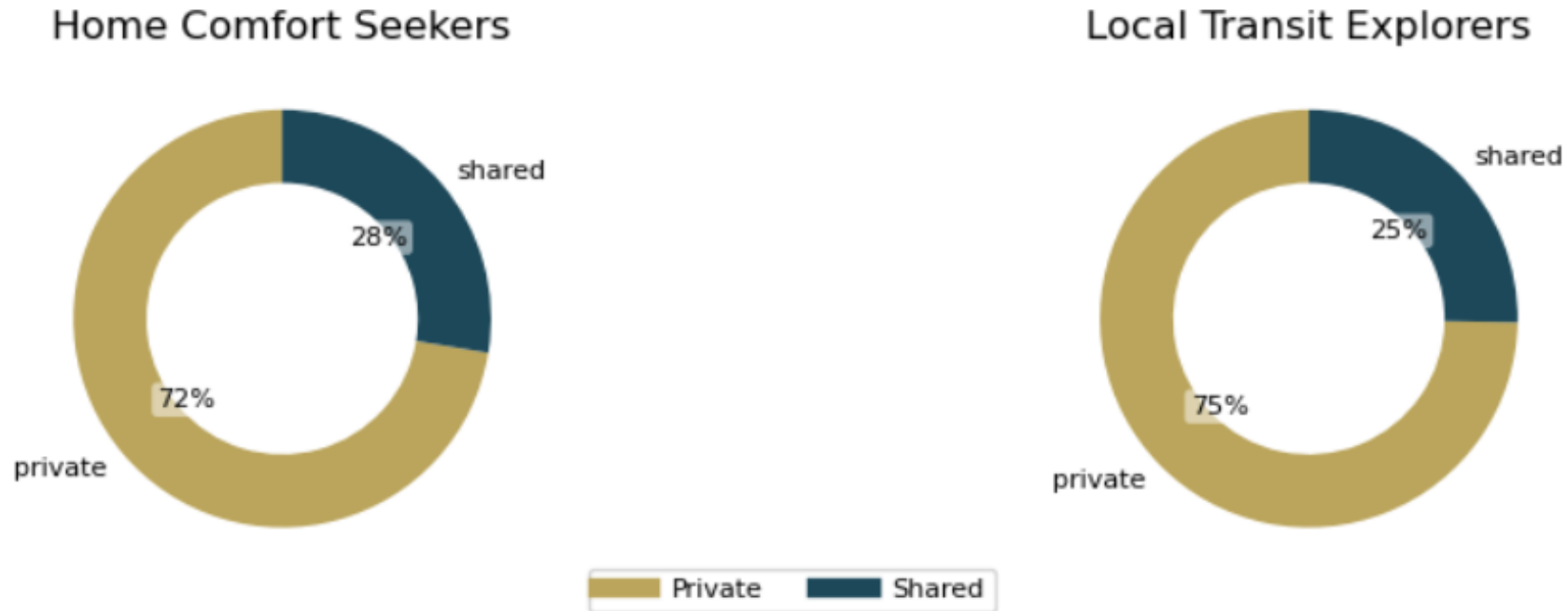
BATHROOMS



A full bathroom is a key requirement for most customers, while half-baths do not significantly influence preferences. Listings should prioritize having at least one complete bathroom.

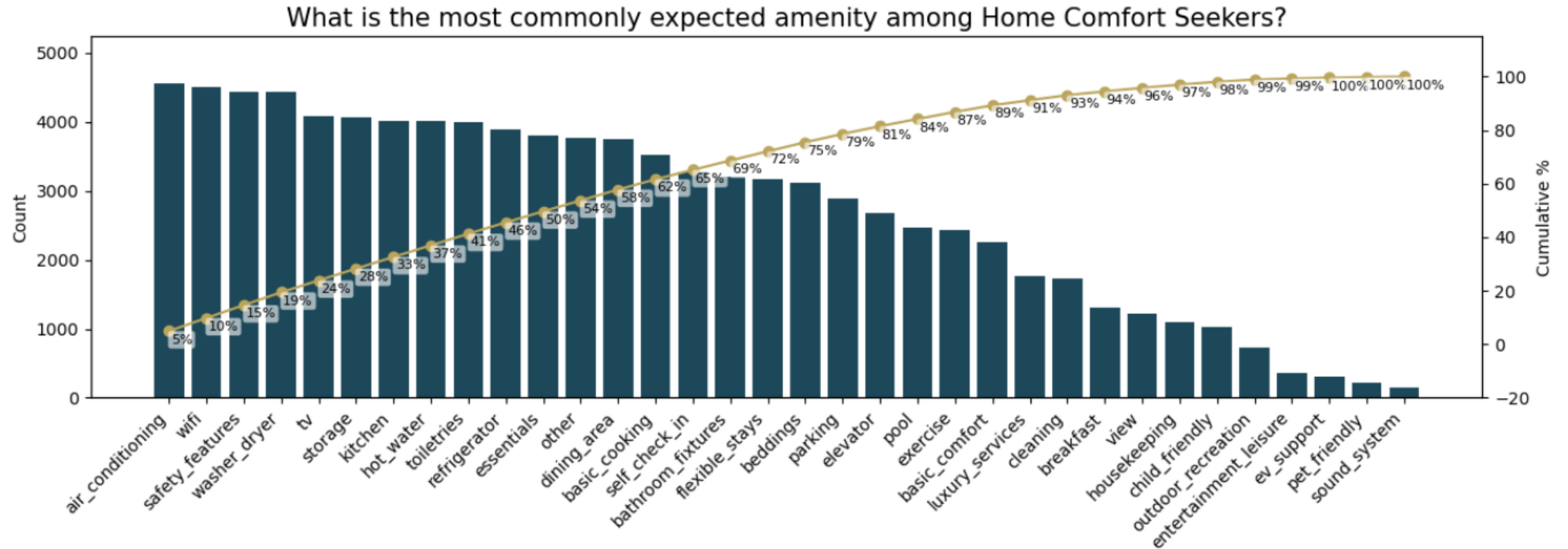
BATHROOM TYPE

Are preferred bathrooms shared or private?



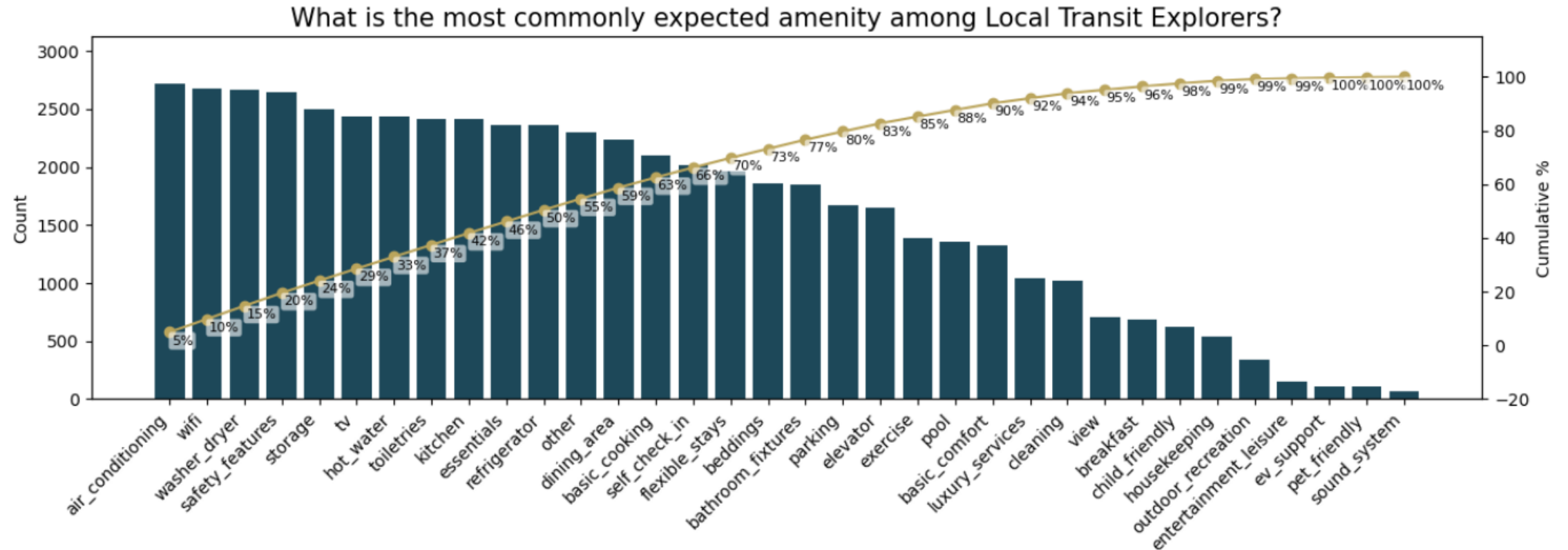
Private bathrooms are strongly preferred by both target customer groups, at about 3x higher than shared bathrooms.

EXPECTED AMENITY



The property should meet at least 80% of guest expectations: air conditioning, wifi, safety features, washer/dryer, tv, storage, kitchen, hot water, toiletries, refrigerator, essentials, dining area, basic cooking, self check-in, bathroom fixtures, flexible stays, beddings, parking, elevator

EXPECTED AMENITY



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PRICING STRATEGIES



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Baseline Price:

- Determined by analyzing the **price distribution** within each target customer segment and identifying the **modal price**, as it reflects the highest demand concentration and helps maximize occupancy.

Guest Surcharges:

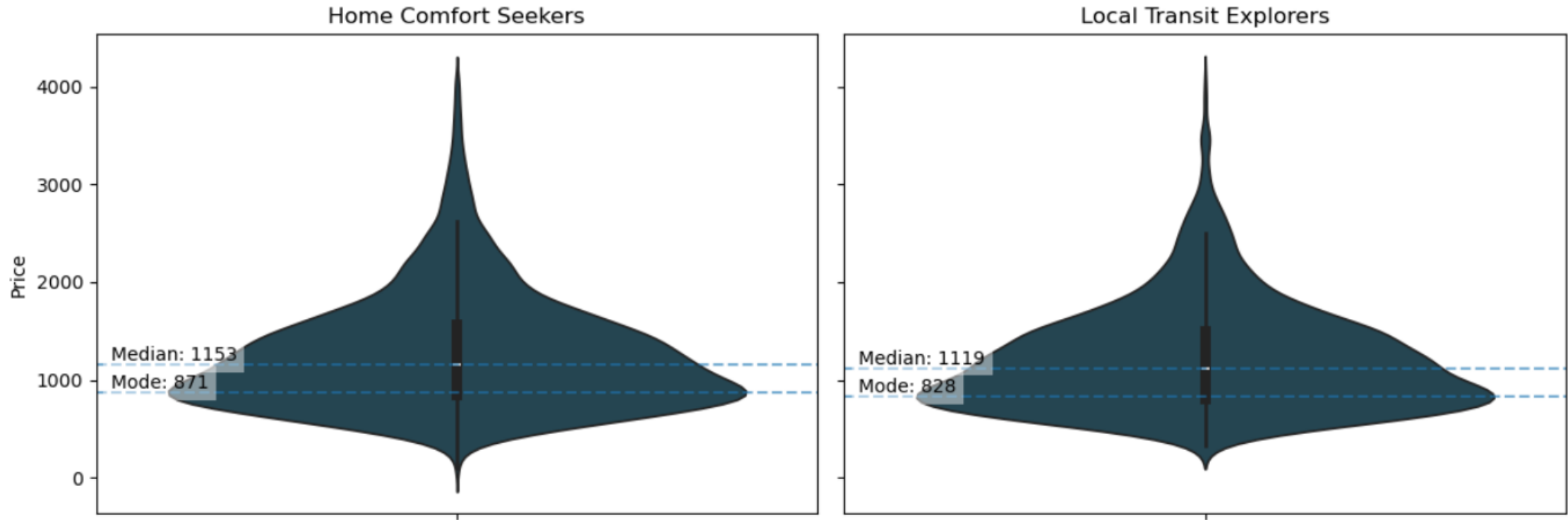
- Estimated by analyzing **median price trends** across listings with different accommodation capacities within each segment, where the price increase per additional guest is derived from the **slope of the overall trend**.

Amenity Premiums:

- Estimated using a **log-linear Ordinary Least Squares (OLS) regression** within each segment, where listing prices (relative to the baseline) are modeled as a function of multiple amenities. **The resulting coefficients** are converted into percentage price adjustments and applied as premiums.

BASE PRICE

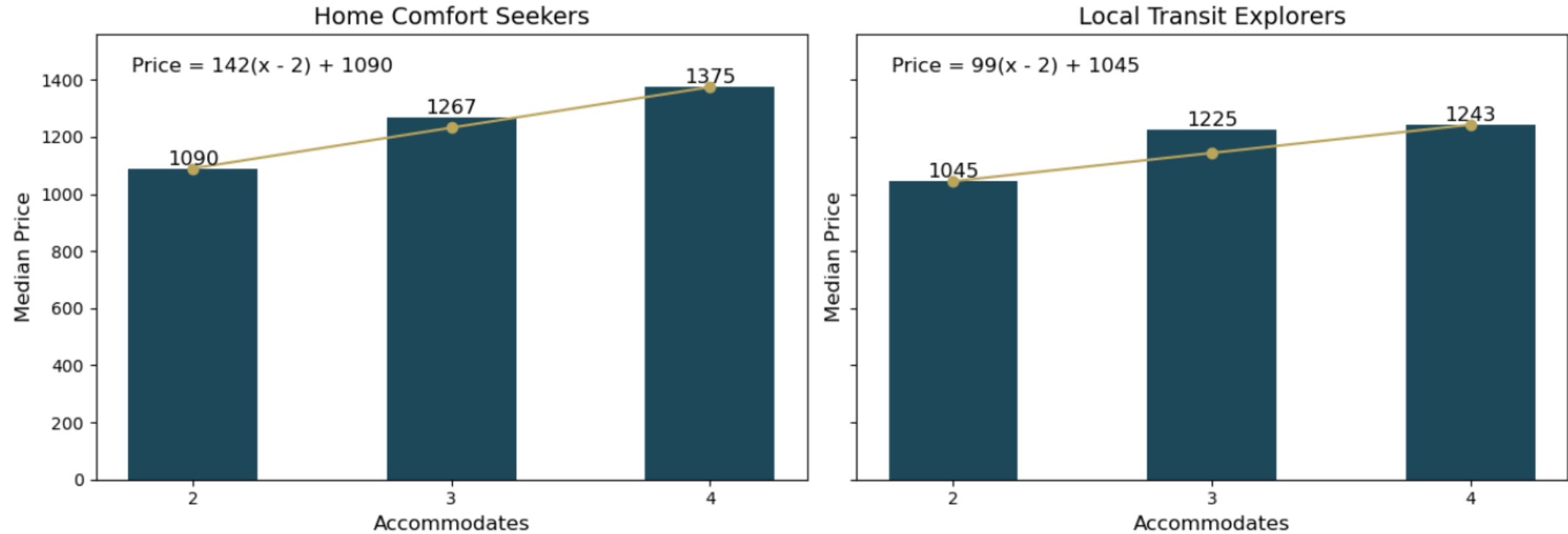
What is the baseline price for each target customer?



Price distribution is similar across both customer segments, with slightly higher values among Home Comfort Seekers. However, demand is most concentrated at lower price points, suggesting that the baseline price should be anchored around the mode to maximize occupancy.

GUEST SURCHARGES

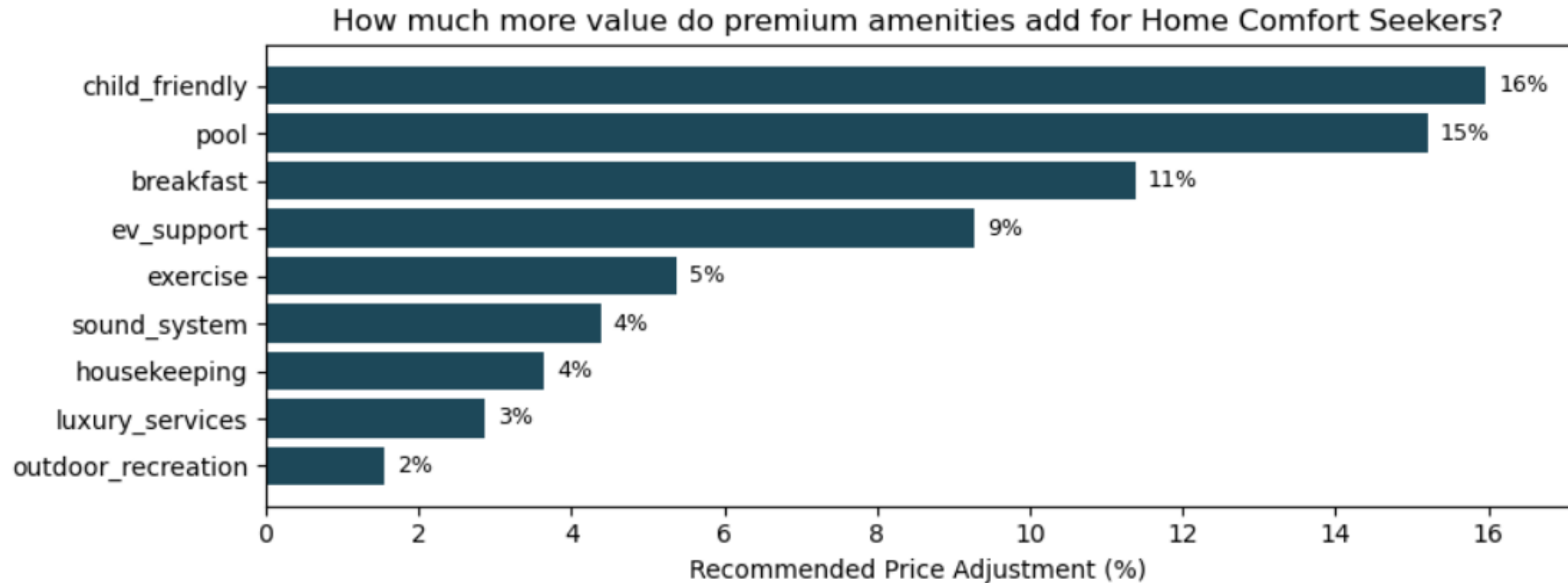
How much more should we charge per extra guest?



Home Comfort Seekers: Recommended 143 THB additional charge per extra guest

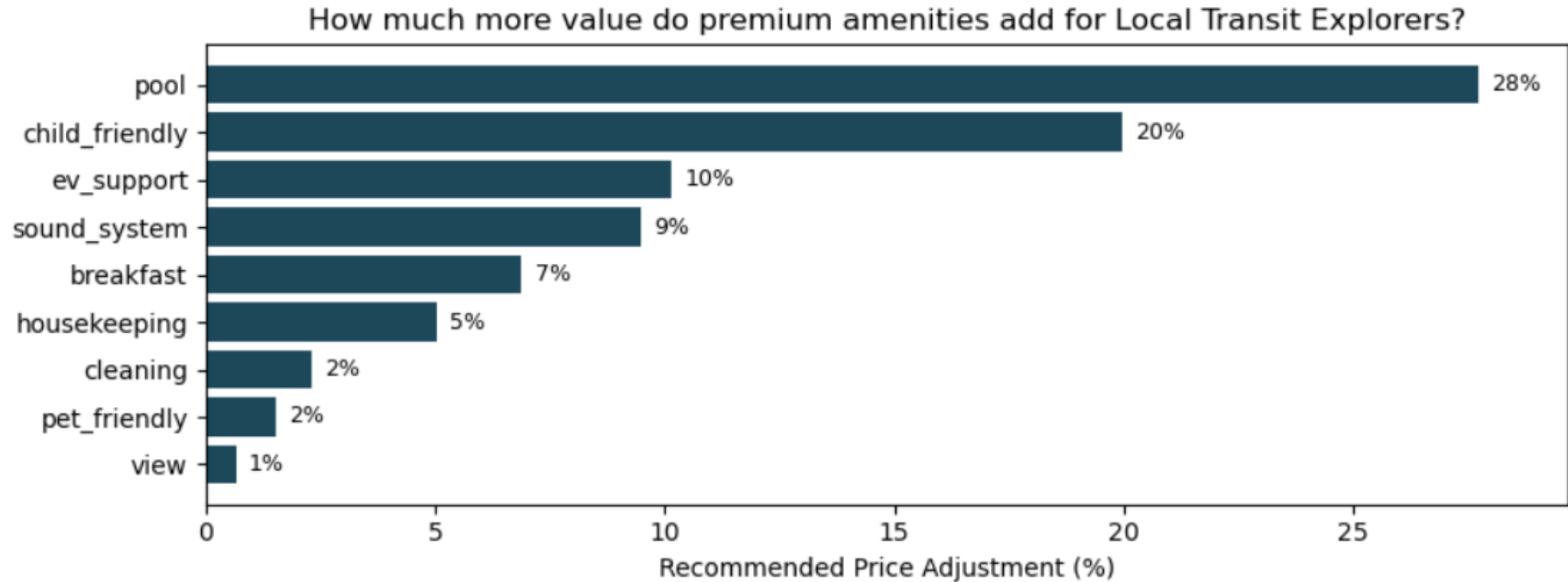
Local Transit Explorers: Recommended 99 THB additional charge per extra guest

AMENITY PREMIUMS



A few key amenities that support family use and everyday comfort, such as child-friendly features, pool, and breakfast, drive most of the pricing impact, while all other amenities have relatively limited influence.

AMENITY PREMIUMS



A single amenity (pool) drives the largest share of pricing influence, followed by child-friendly features, indicating a more concentrated, leisure-oriented preference. All other amenities contribute significantly less.